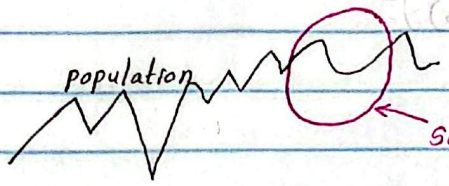
No. .....

# Linear Regression

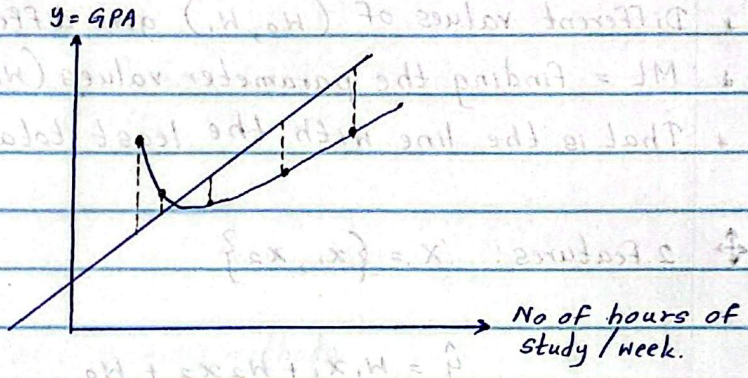
All models are wrong. Some are useful.

Curve Fitting → Finding a mathematical function or model that represents the relationship between input data and output data.




Training data

| x | y |
|---|---|
| 5 | 2.7 |
| 2 | 3.2 |
| 1 | 4 |
| 3 | 2.9 |
| 6 | 3.5 |




Linear regression uses a linear function to fit the data.

$$\hat{y} = mx + c$$

$$E = \sum_{i=1}^{5} (\hat{y}_i - y_i)^2$$

$$\hat{y}_i = mx_i + c$$

$$E = \sum_{i=1}^{5} [(m_i x + c_i) - y_i]^2$$

$$E = f(m, c)$$

The error depends on the values of m and c.

Bias-variance problem is an issue in curve fitting.

- High bias - model is too simple → under fitting
- High variance - model is too complex → overfitting
- Good model → balaces both.

Bias variance problem = Finding the right model complexity.

Hyperplane - A decision boundry that seperates data into different classes.

EDUCATION IS THE BEST INVESTMENT

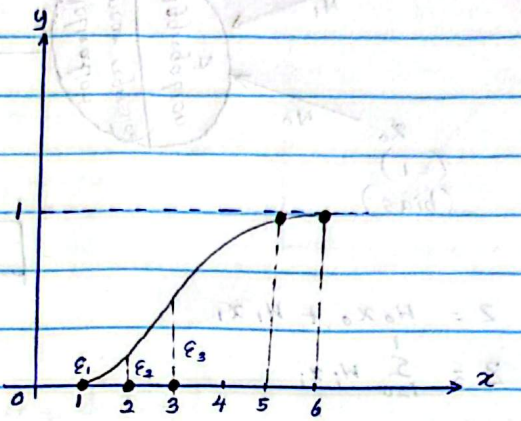
CamScanner

No. Date.

# Logistic Regression.

## Classification problem

| x | y |
|---|---|
| 2 | 0 |
| 1 | 0 |
| 5 | 1 |
| 3 | 0 |
| 6 | 1 |
| 4 | ? |




Instead of directly using a straight line as the output logistic regression uses a sigmoid function.

Sigmoid Function → $\hat{y} = \frac{1}{1+e^{-(mx+c)}}$ ; $0 \le \hat{y} \le 1$

$$E = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

$$E = \sum_{i=1}^{n} \left[ y_i - \frac{1}{1+e^{-(mx+c)}} \right]^2$$

$$E = F(m, c)$$

## With multiple inputs

(Model A) - Linear regression model : $\hat{y} = \overset{W_1}{m_1} x_1 + \overset{W_2}{m_2} x_2 + \overset{W_0}{c}$

$$\underline{\hat{y} = \sum_{i=0}^{2} W_i x_i} \quad ; \; x_0 = 1$$

(Model B) - Logistic regression model : $\hat{y} = \frac{1}{1+e^{-(\overset{W_1}{m_1} x_1 + \overset{W_2}{m_2} x_2 + \overset{W_0}{c})}}$

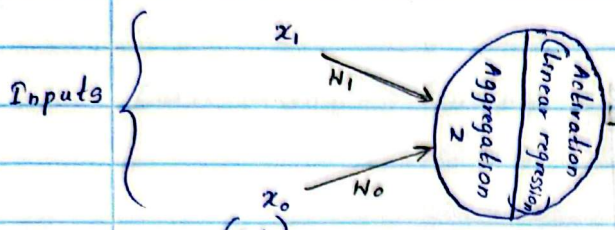
$$\underline{\hat{y} = \frac{1}{1+e^{-\left(\sum_{i=0}^{2} W_i x_i\right)}} \quad ; \; x_0 = 1}$$

EDUCATION IS THE BEST INVESTMENT

CamScanner

No. .....

## Single Neuron Architecture.




$\hat{y} = z$

$\hat{y} = \sum_{i=0}^{2} H_i x_i$

$$\hat{y} = \underbrace{H_0 x_0}_{c} + \underbrace{H_1 x_1}_{m}$$

$z = H_0 x_0 + H_1 x_1$

$z = \sum_{i=0}^{1} H_i x_i$

Single neuron based thinking.

| $x_1$ | $y$ |
|---|---|
| 2 | 2.8 |
| 1 | 3.2 |
| 3 | 3.8 |
| 5 | 3.6 |
| 6 | 2.9 |

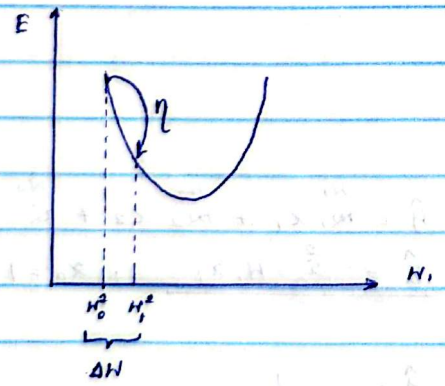
If $H_1 = 0.8$, $H_0 = 1.2$

- $2 \times 0.8 + 1 \times 1.2 = z = \hat{y} = [2.8 - 2.8] = 0$
- $1 \times 0.8 + 1 \times 1.2 = z = \hat{y} = [2.0 - 3.2] = (1.2)^2$

$\vdots$

$E$

Weight learning - Final output minimizing the error.




$$H_1^2 = H_1' + \Delta H_1'$$

$$H_0^2 = H_0' + \Delta H_0'$$

Epoch - one complete exposure / pass through the entire training dataset.

Ex: If the data set contains 1000 samples, model processes all 1000 once : 1 epoch.

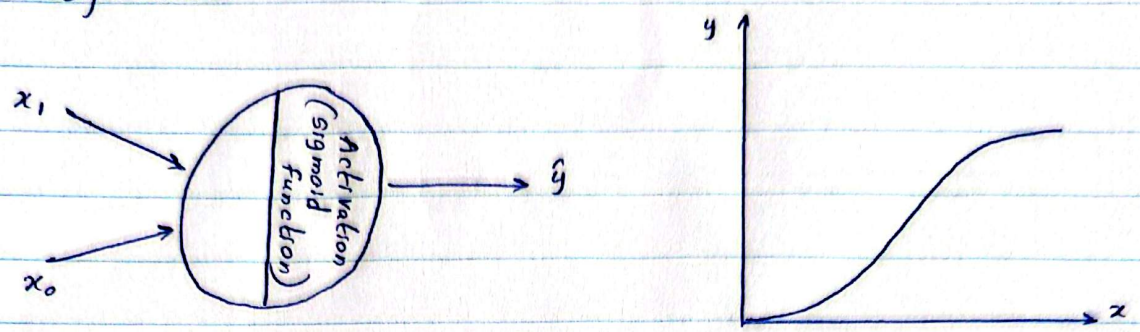
Why multiple epochs are used → A model may not learn enough from seeing the dataset only once.

EDUCATION IS THE BEST INVESTMENT

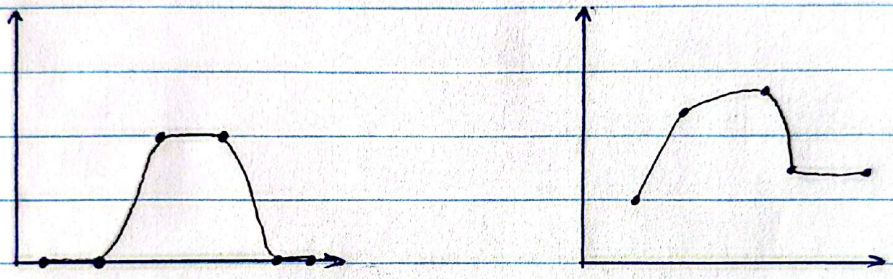
CamScanner

No. .....  Date. .....

Classification problem with single neuron architecture is equal to logistic regression.




Simple models cannot fit every dataset.



Logistics  Linear.

These datasets with shapes cannot easily be represented using only a straight line or basic sigmoid curve.

* Model that is too simple to capture the pattern - Underfitting
* Model becomes unnecessarily complex and memorizes training data - Overfitting

CamScanner